

Oil & Gas Predictive Maintenance using Machine Learning

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Domain: Data Science & Machine Learning

1. Project Overview

This project develops a machine learning solution to predict equipment failures in the Oil & Gas industry using historical sensor readings. The solution covers data preprocessing, exploratory data analysis (EDA), model training, evaluation, and deployment using Streamlit.

2. Problem Statement

Unexpected equipment failures result in production downtime, increased maintenance costs, and safety risks. The objective is to predict failures before they occur so maintenance can be scheduled proactively.

3. Objectives

- Predict equipment failures using sensor data.
- Compare multiple machine learning algorithms.
- Deploy the best model as an interactive web application.

4. Dataset Description

Dataset size: 5,025 records. Target: Failure (0 = No Failure, 1 = Failure). Features include temperature, pressure, vibration, flow rate, motor current, oil level, runtime hours, ambient temperature, humidity, maintenance days, previous failures and equipment type.

5. Data Preprocessing

Missing value analysis, duplicate check, one-hot encoding, feature scaling, train-test split and preparation for model training.

6. Exploratory Data Analysis

Performed class distribution analysis, correlation heatmap, boxplots, histograms and feature distribution analysis to understand the dataset.

7. Machine Learning Models

Logistic Regression, Random Forest and XGBoost classifiers were trained and evaluated.

8. Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	78.01%	20.00%	91.53%	32.83%	84.00%
Random Forest	91.34%	31.58%	40.68%	35.56%	67.59%
XGBoost	94.03%	46.15%	10.17%	16.67%	54.71%

9. Best Model Selection

Logistic Regression achieved the highest Recall (91.53%) and ROC-AUC (84%), making it suitable when detecting equipment failures is the highest priority.

10. Streamlit Deployment

An interactive Streamlit application was developed to accept sensor inputs, predict equipment failure, display prediction probability and provide maintenance recommendations.

11. Future Scope

Use larger industrial datasets, integrate IoT sensors, apply SHAP explainability, optimize hyperparameters and deploy on cloud infrastructure.

12. Conclusion

The project demonstrates a complete end-to-end machine learning workflow for predictive maintenance. It can support proactive maintenance decisions and reduce unexpected equipment downtime.